MATERIAL CERTIFICATION REPORT - FORGINGS, BARS, AND MILL SHAPES

CUSTOMER:	CACTUS PO NO.	CACTUS LINE TEM NO.
CACTUS WELLHEAD(SUZHOU)PRESSURE CONTROL CO.,LTD.	SUP0025187	Line2

DATE:	PACKING LIST NO.	MATERIAL SPEC/REVISION:	MATERIAL SPEC/REVISION:
2024/6/14		CMS-107 H	CMS-002, REV K

The material chemistry, heat treatment and mechanical testing was transcribed from the following manufacturer's certificate of tests.

MANUFACTURER ONE:	MANUFACTURER TWO:	MANUFACTURER THREE:
JIANGSU JINGYI FORGING CO., LTD	Jinhu Haiyuan Machinery Co., Ltd	SUZHOUXINSU PHYSICAL&CHEMICAL TESTING SERVICE CO., LTD

QTY:49	PART NUMBER/REVISION: 134797 A02	DESCRIPTION/SIZE:CSGHGR,CW,MBU-3T-TP8-UPR,SN,7-5/8,FLUTED,11 NESTED W/4.0 ID X 6.5 OD X BLANK PIN BTM X 30.5 LG X6.125-4 STUB ACME-2G RIGHT HAND BOX TOP & 4-13/16 HBPV THD,SPEC FOR ROTATING CASING STRING,4140 125K,TEMP U,MATL AA,PSL3,PR2

MELT PRACTICE:	FORGING RATIO:	MATERIAL GRADE:	GRAIN SIZE:	IDEAL CRITICAL DIAMETER (DI):
BOF+LF+VD+CCM	10.31:1	4140	7.5	

HEAT NUMBER:	HEAT TREAT LOT NUMBER:	SERIAL NUMBERS:
2310345	HY240129-2	SUP0025187-2-52~SUP0025187-2-100

CHEMICAL ANALYSIS

%C	%Mn	%P	%S	%Si	%Cu	%Ni	%Cr	%Mo	%V	%AL	%NB+TA	%Nb
0.393	0.911	0.010	0.0048	0.207	0.011	0.143	0.999	0.153	0.0051	/	/	/

MECHANICAL PROPERTIES

ASTM A-370	REQUIREMENT	ACTUAL
Tensile Strength (PSI) (.505 DIA)	⩾1000mPa	1175
.2% Offset Yield Strength (PSI)	⩾862mPa	1095
% Elongation in 2" Gage Length	⩾12%	19
% Reduction in Area	⩾30%	56
Surface Hardness (BHW)	/	/

CHARPY IMPACT PROPERTIES

10MM x 10MM	ACTUAL	AVERAGE
Foot Pounds/Joules	40 31 33	34.67
CVN Orientation	☐ Transverse	☑Longitudinal
Temperature (°F OR °C)	-18℃	
QTC Hardness (Two Brinell Readings): 311 319		
QTC Size/ Part: :Ontologyto extend the test block		

HEAT TREATMENT

PROCESS STEP	TEMPERATURE	HOURS	QUENCH MEDIA	QUENCH BATH TEMP (BEFORE & AFTER)
Normalize	900℃	4.0h	Air	B: A:
Austenitize-Solution Annealed	875℃	4.1h	PAG	B: 22℃ A :25℃
First Temper-Age	620℃	6.0h	Air	
Second Temper -Age				

QTC HEAT TREATMENT

TEMPERATURE	HOURS	QUENCH MEDIA	QUENCH BATH TEMP (BEFORE & AFTER)
900℃	4.0h	Air	B: A:
875℃	4.1h	PAG	B: 22℃ A 25℃
620℃	6.0h	Air	

BARSTOCK HARDNESS PROFILE

HARDNESS OD: HB 293-321	HARDNESS CENTER: HB 293-321	HARDNESS MID-RADIUS: HB 293-321

PLEASE CHECK THE BOXES IF APPLICABLE:

☑MATERIAL IS CAPABLE OF MEETING THE SURFACE AND VOLUMETRIC NDE OF API SPEC 6A

☑ NO WELDING HAS BEEN PERFORMED ON THIS MATERIAL

☑ FURNACES CALIBRATED AND CERTIFIED TO API SPEC 6A, ANNEX M FOR BATCH-TYPE AND SAE AMS-H-6875 FOR CONTINUOUS-TYPE

We hereby certify that the raw material data above is a true copy of the data furnished by the original manufacturer/mill/testing laboratory and complies with the terms of the Cactus Wellhead Purchase Order. Original manufacturer/mill/testing laboratory certificate(s) meeting EN 10204 Type 3.1 have been provided.

CERTIFIED BY: Jiubenzhu	POSITION TITLE:Checker

Form CMS-002 Rev D

江苏精异锻造股份有限公司
Jiangsu JingYi forging co., LTD
地址：江苏省阜宁县陈集工业园 218 号
Address:ChenJi zhen Funing County Industrial Park No.218
Telephone:0515-87838882 Fax:0515-87838887

锻造报告
Forging Report

Document No. JSJY/QR-JZ-10 (Rev. A/1) Date: 2023.12.23

客户 Client	苏州派普机械/Suzhou Pympe Machinery CO.,Ltd		报告编号 Report No.	2023372
	采购订单 Purchasing Order	PP-W020230098	生产单号 production Order	/
订单要求 PO Requirements	产品规格 Description	CSGHGR,CW,MBU-3T-TP8-UPR,SN,7-5/8,FLUTED,11 NESTED W/4.0 ID X 6.5 OD X BLANK PIN BTM X 30.5 LG X 6.125-4 STUB ACME -2G RIGHT HAND BOX TOP &4-13/16 HBPV THD SPEC FOR ROTATING CASING STRING,4140 125K,TEMP U,MATLAA,PSL3 ,PR2	零件号 Part Number	134797 A02
	材料规范 Material Spec.	CMS-107 H	材料牌号 Material Grade	AISI 4140
	产品数量 Quantity	135	试块数量 QTC Quantity	3
	试块尺寸 QTC Size	本体延长	产品序列号 Serial NO.	SUP0025187-1-67~100 SUP0025187-2-1~101
原材料 Raw Material	类型 Tyoe	■ 棒料 Bar □ 连铸 Con.Cast □ 方坯 Billet		□ 钢锭 Ingot
	毛坯尺寸 Size	Ø200mm	钢厂炉号 Heat No	2310345
	熔炼方法 Melt Practice	BOF+LF+VD+CCM	晶粒度 Grain Size	7.5
	钢厂 Mill Plant	芜湖新兴铸管		
锻造 Forging	设备型号 Equipment Type	6000T Hydraulic Press	毛坯重量 Weight	260kg
	始锻温度 Initial Forging Temperature	1150°C	终锻温度 Final Forging Temperature	850°C
	锻造比 Forging Ratio	钢厂锻造比 Forging Ratio from steel mill	锻造厂锻造比 Forging Ratio from forging supplier	最终锻造比 Final Forging Ratio
		9:1	1.31:1	10.31:1

化学分析结果(重量百分比)Chemical Analysis Results(Weight%)

C%	Si%	Mn%	P%	S%	Ni%	Cr%	Cu%	Mo%	V%
0.38-0.43	0.15-0.32	0.75-1.00	0.025Max	0.025Max	0.50Max	0.8-1.10	-	0.15-0.25	0.05Max
0.420	0.251	0.998	0.013	0.002	0.030	1.010	0.020	0.221	0.005

机械性能 Mechanical Properties

屈服强度 Yield Strength	抗拉强度 Tensile Strength	延伸率 Elongation	断面收缩率 Reduction of Area	硬度值 Hardness(HB)	冲击试验(温度：) Impact Test(Temp:)
/	/	/	/	/	/

锻后热处理 Heat Treatment after Forged

过程 Process	加热炉类型 Furnace Type	保温温度 Holding Temp.	保温时间 Holding Time	冷却介质 Cooling Media
□ 正火 Normalize	□ 电 Electric □ 气 Gas □ 煤 Coal	/	/	/
□ 淬火 Quench	□ 电 Electric □ 气 Gas □ 煤 Coal	/	/	/
□ 回火 Temper	□ 电 Electric □ 气 Gas □ 煤 Coal	/	/	/

注 Note：淬火前后介质温度 Batch Temperature Before: After:

检验和测试 Inspection and Test

方法 Method	■ 尺寸和外观 Dim & Visual	□ 超声 UT	□ 磁粉 MT	□ 着色 PT
结果 Results	合格 Accept			
其它 Other				

兹证明材料已按材料规范和采购订单的要求进行了制造、检验和试验，结果符合要求。
We hereby certify that the material have been manufactured, inspected and tested in accordance with the material specification and the purchase order requirements, and found to meet the requirements.

质量部签名 Quality Manager(Signature):

热处理报告
Heat Treatment Report

热处理供应商	金湖海源机械有限公司 jinhu haiyuan Machinery Co., Ltd			
客户 Client	Suzhou Pympe Machinery CO.,Ltd			
	采购订单 Purchasing Order	PP-W020230098		
产品 Component	零件号 Part Number	134797 A02	图号 Drawing No.	MD12844 REV. B
	零件描述 Part Description	CSGHGR,CW,MBU-3T-TP8-UPR,SN,7-5/8,FLUTED,11 NESTED W/4.0 ID X 6.5 OD X BLANK PIN BTM X 30.5 LG X6.125-4 STUB ACME-2G RIGHT HAND BOX TOP & 4-13/16 HBPV THD,SPEC FOR ROTATING CASING STRING,4140 125K,TEMP U,MATL AA,PSL3,PR2		
	材料牌号 Material Designation	4140	数量 Quantity	50
	冶炼炉号 Heat No.	2310345	追踪号 Tracking No.	SUP0025187-2-52~101
	热处理工艺规范 HT Procedure	CMS-107H	热处理批次 HT Lot No.	HY240129-2

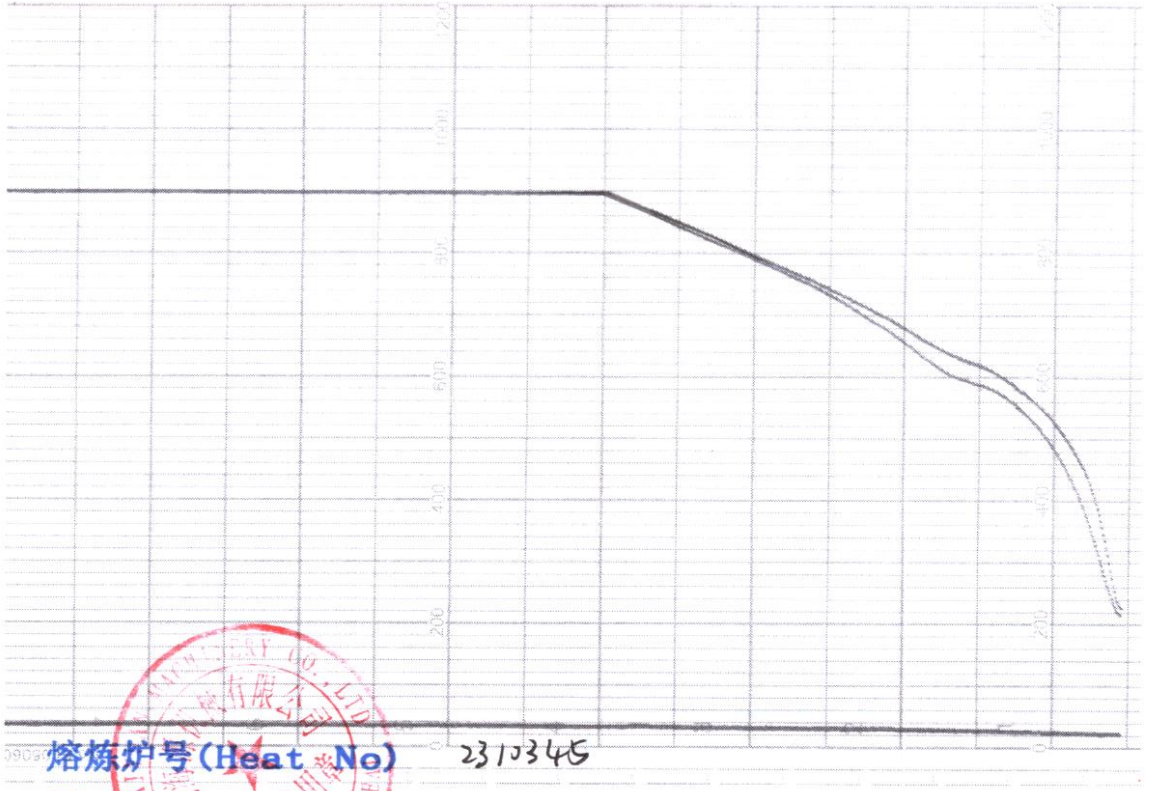
热处理过程 Heat Treatment Process

项目名称 Item	工件热处理 Component Treatment			鉴定试块热处理 QTC Heat Treatment		
	正火 Normalize	奥氏体华/淬火 Austenitize/ quench	回火 Temper	正火 Normalize	奥氏体华/淬火 Austenitize/ quench	回火 Temper
进炉温度 Initial Temperature	220℃	320℃	100℃	220℃	320℃	100℃
加热速率 Heating up rate	194℃/h	140℃/h	260℃/h	194℃/h	140℃/h	260℃/h
保持温 Holding temperature	900℃	875℃	620℃	900℃	875℃	620℃
保温时间 Holding time	4.0H	4.1H	6.0H	4.0H	4.1H	6.0H
冷却介质 Cooling Media	Air	PAG	Air	Air	PAG	Air
热处理炉编号 HT Furnace No.	HY-Z-06 RZJ3-360-12	HY-Z-06 RZJ3-360-12	HY-Z-08 RZJ3-210-7	HY-Z-06 RZJ3-360-12	HY-Z-06 RZJ3-360-12	HY-Z-08 RZJ3-210-7
炉温均匀性曲线校准期 Furnace Survey Calibration Date	2023.06.20 2024.06.19	2023.06.20 2024.06.19	2023.06.21 2024.06.20	2023.06.20 2024.06.19	2023.06.20 2024.06.19	2023.06.21 2024.06.20
温控仪校准期 Temperature controller Calibration Date	2023.12.07 2024.03.06	2023.12.07 2024.03.06	2023.12.07 2024.03.06	2023.12.07 2024.03.06	2023.12.07 2024.03.06	2023.12.07 2024.03.06
记录仪校准期 Recorder Calibration Date	2023.12.07 2024.03.06	2023.12.07 2024.03.06	2023.12.07 2024.03.06	2023.12.07 2024.03.06	2023.12.07 2024.03.06	2023.12.07 2024.03.06
热电偶校准期 Thermocouple Calibration Date	2023.12.07 2024.03.06	2023.12.07 2024.03.06	2023.12.07 2024.03.06	2023.12.07 2024.03.06	2023.12.07 2024.03.06	2023.12.07 2024.03.06

注 Note:淬火开始时,PAG 淬介质的温度不得超过 38℃并且在淬火循环中的任何时候均不得超过 49℃
The temperature of the PAG shall not exceed 38℃ and the start of the quench. For bath-typequenching,
The temperature of the PAG shall not exceed 49℃ at any time during the quench cycle.
淬前介质温度 Bath Temperature Before: 22℃　　淬火过程中介质温度 Bath Temperature During 30℃
淬后介质温度 Bath Temperature After: 25℃

兹证明材料已按材料规范和采购订单的要求进行了热处理、检验，结果符合要求。
We hereby certify that the material have been heat treated and inspected, in accordance with the material specification and the purchase order requirement, and found to meet the requirements.

编制 Prepared by: 房有林 24.2.1　　审核 Audited by:柏仲海 24.2.1



熔炼炉号(Heat No) 2310345

工　艺(Technical) N

热批号(Heat Treatment Lot No) 10/240129-2

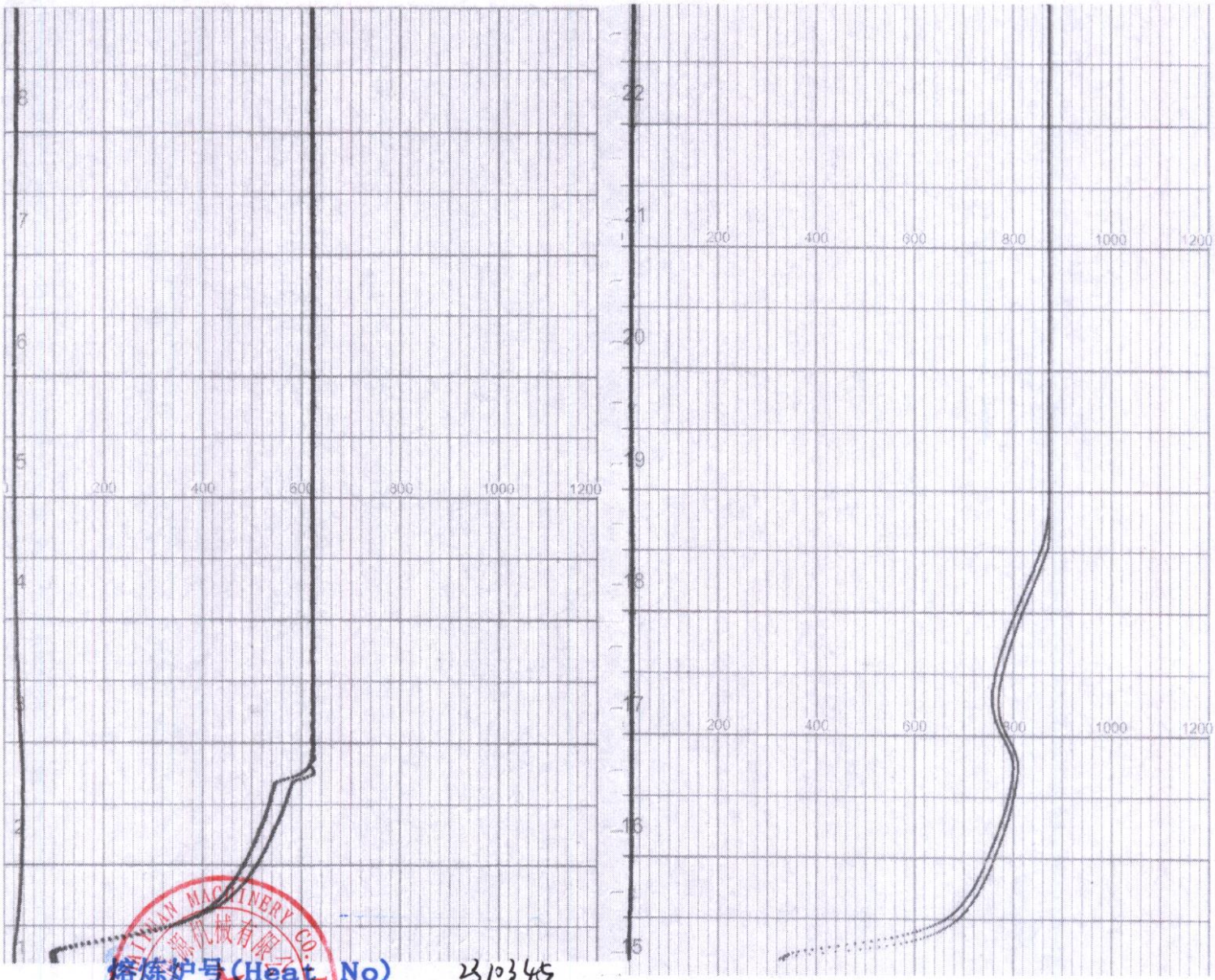
日　期(Date) 2024-1-29

材　料(Material) 4140

操作者(Operator) 彦林林

清　单(Description)

134797 1002 MD12844 REV.B 50件



熔炼炉号(Heat No) 23103-45

工　艺(Technical) 8PT

热批号(Heat Treatment Lot No) 1042401292

日　期(Date) 2024-1-30

材　料(Material) 4140

操作者(Operator) 彭振林

清　单(Description)

134797 1002 MD12844 REV: B 50件

苏州新苏理化测试服务有限公司

SuZhou Xinsu Physical&chemical Testing Service Co., Ltd

检 测 报 告
TEST REPORT

委托单位： Client：	苏州派普机械有限公司 Pympe Machinrey Co.,Ltd.
样品名称： Sample Name：	材质 Material：4140，炉号 Heat No.：2310345 热处理批次号 Heat Treat Lot No.：HY240129-2 试块大小 QTC size：Ontologyto extend the test block
样品描述： Sample Description：	圆棒拉伸试样 1 件，10×10×55V 型冲击试样 3 件（1#~3#），硬度试样 1 件。 1 piece of round tensile specimen，3 pieces of impact coupons with type of 10×10×55V（1#~3#），1 piece of hardness specimen.
收样日期： Date of receipt：	2024-03-04
报告日期： Date of report：	2024-03-05

1.本报告无本公司检测专用章无效。Test report without the company's inspection seal is invalid.
2.本结果仅对来样负责。This result is only responsible for the sample.
3.对检测报告有异议在 15 日之内提出，逾期不以受理。If there is objection about the report, indicate within 15 days, overdue not be with accepted.
4.未经允许本报告不得部分复印。This report cann't be copied without the permission.
5.相关项目未取得 CMA 资质认定，仅作为内部质量控制之用。Relevant items not accredited by CMA, only for internal quality control.

编　制： Editor：	张　玉	张玉
审　核： Approver：	文悦莲	文悦莲
签　发： Issuer：	徐卫龙	徐卫龙
签发日期： The issue date：	2024-03-05	

苏州新苏理化测试服务有限公司 检测专用章

地址/Add：苏州市高新区嵩山路 252 号 7 幢/ No. 252-7, Songshan Road, High-Tech Zone, Suzhou
邮编/P.C.：215129　　电话/Tel/传真/Fax：0512-65579663　　E-mail：xslhcs@126.com

检 测 报 告
Test Report

拉伸试验 Tensile Test	
检测方法 Test Methods：	ASTM A370-23
设备 Equipment：	WEW-300B 液压式万能试验机（编号：XS0901） Hydraulic universal material testing machine（No.：XS0901）

检测项目 Test Item	单位 Unit	范围 Range	结 果 Result
抗拉强度（Su） Tensile strength	MPa	1000 Minimum	1175
屈服强度($R_{p0.2}$) YS(offset=0.2%)	MPa	862 Minimum	1095
断后伸长率（A） Percentage elongation after fracture	%	12 Minimum	19
断面收缩率（Z） Percentage reduction of area	%	30 Minimum	56

备注/Note：样本尺寸 Specimen size:Φ12.50mm；
Test result meets the requirements of CMS-107 RevH 测试结果符合 CMS-107 版本 H 的要求。
温度 Temperature：23℃，湿度 Moisture：53%RH

冲击试验 Impact Test	
检测方法 Test Methods：	ASTM A370-23
设备 Equipment：	JB-300B 冲击试验机（编号：XS2201） JB-300B Impact Testing Machine（No.：XS2201） CDW-80 冲击试验低温槽（编号：XS2003） CDW-80 Impact test at low temperature trough（No.：XS2003）

检测项目 Test Item	试样尺寸(mm) Specimen Dimensions	单位 Unit	纵向测试结果 longitudinal Direction Result		
			1#	2#	3#
-18℃（0°F）冲击吸收能量(KV_8) Impact Absorbed Energy	10×10×55	J	40	31	33

备注/Note：
1.冲击试验机打击能量 300J/Impact Testing Max. impact energy 300J
2.Test result meets the requirements of CMS107 Rev H 测试结果符合 CMS107 版本 H 的要求。
3.范围 Range: 10×10×55 冲击功平均值（AVE）：$KV_8 \geq 20$，允许有 1 个：$14 \leq KV_8 < 20$
10×10×55 Average Impact Value (Average)：$KV_8 \geq 20$，only one result allowed within $14 \leq KV_8 < 20$
温度 Temperature：23℃，湿度 Moisture：53%RH

接下页
To be continued

苏州新尚理化测试服务有限公司 检测专用章

检 测 报 告
Test Report

布氏硬度试验 Brinell Hardness Test	
检测方法 Test Methods：	ASTM E10-23
设备 Equipment：	HB3000 布氏硬度计（编号：XS1904） HB3000 Brinell Hardness Tester（No.：XS1904）

检测项目 Test Item	范围 Range	结 果 Result		
布氏硬度（HBW10/3000） Brinell Hardness	293~327	311	319	317
备注/Note：Test result meets the requirements of CMS-107 RevH 测试结果符合 CMS-107 版本 H 的要求。 温度 Temperature：23℃，湿度 Moisture：53%RH				

以下空白
End of report

地址/Add：苏州市高新区嵩山路 252 号 7 幢/ No. 252-7, Songshan Road, High-Tech Zone, Suzhou
邮编/P.C.：215129 电话/Tel/传真/Fax：0512-65579663 E-mail：xslhcs@126.com

Ultrasonic Inspection Report

CACTUS JOB#: N/A CACTUS PO#: SUP0025187 WO #: / DATE: 2024-02-18

LOCATION/BRANCH: Suzhou Pympe Machinery CO.,Ltd MATERIALOR TYPE: 4130 75K

EQUIPMENT DESCRIPTION: CTS-2020 PART NUMBER/REV: 134797/A02 MATERIALSIZE/THICKNESS: Ø282*783

CUSTOMER PRODUCT IDENTIFIER: SUP0025187-2-76~SUP0025187-2-100

INSPECTION PROCEDURE

NDT SPECIFICATION CNT-005,UltrasonicTesting REVISIONE E

Cactus Wellhead NDT Specification contains inspection methodology and acceptance criteria

MODE	☐ SHEAR ☒ LONGITUDINAL	SCANNING METHOD	☒ CONTACT ☒ MANUAL ☐ IMMERSION ☐ AUTOMATIC	PRES. TYPE	☒ A-SCAN ☐ PHASEDARRAY ☐ DIGITALTHICKNESS	AREAS EXAMINED	☒ ALL ACCESSIBLEVOLUME ☐ PRESSURECONTAININGWELD(S) ☐ CLADWELD(S) ☐ OTHER(EXPLAINBELOW)

SURFACE CONDITION: After HT ≤250Rms MATERIAL THICKNESS/DIA: Ø282*783

UT INSTRUMENTMAKE/MODEL: CTS-2020 S/N AND CAL. DUE DATE: 526322181062/2024-09-05

COUPLANT: Machine Oil ANGLE: 0°

CABLETYPE/LENGTH: Q9-Q9,Double Q9-C5/1.8m TRANSDUCERMANUFACTURER: Ultrabetter Testing Technology (Suzhou) Co.LTD

TRANSDUCERSIZE: Φ20/Φ14/Φ10 TRANSDUCERFREQUENCY: 2.5MHZ/5.0MHZ

REFERENCE DB: 80% RECORD ADDITIONAL DB'S FOR: CURVATURE: 29dB ACOUSTICDIFFERENCES: 4dB SCANNING: +6dB

(TOTAL): P1-65dB/P2-52dB/P3-62dB/P4-53dB

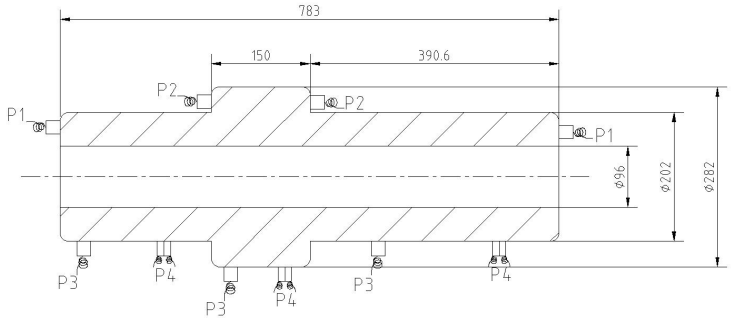
REFERENCE STANDARDS

BLOCK SERIAL NUMBER	OVERALL THICKNESS	METAL TRAVEL DISTANCE	REFLECTOR SIZE	REFLECTORORIENTATION
P1:4130-16-0787/4130-16-2362/4130-16-3150	800mm	783mm	1/4"	FBH/straight(2.5Z20N)
P2:4130-8-0099/4130-8-0296/4130-8-0600	152.4mm	150mm	1/8"	FBH/straight(5Z14N)
P3:4130-8-0099/4130-8-0197/4130-8-0394	100mm	93/53mm	1/8"	FBH/straight(5Z14N)
P4:4130-8-0025/4130-8-0059/4130-8-0099	25mm	93/53mm	1/8"	FBH/Double straight(DA5P10FS25)

INSPECTION RESULTS

WORK ORDER&SERIAL #	HEAT #	HEAT LOT #	SERIAL#(ORIGINAL)	ACCEPT/REJECT QUANTITIES	NCR #(IF APPLICABLE)
/	2310345	HY240129-2	SUP0025187-2-76~SUP0025187-2-100	25 ACCEPT	NA

SKETCH AND TECHNIQUE DESCRIPTION (Identify and attach additional drawings/sheets):



MIN. THICKNESS REQUIRED:	N/A	MIN. THICKNESS RECORDED:	N/A	ACCEPT OR REJECT:	Accept all		
INSPECTION PERFORMED BY:	Sunpengyang CO:21102429098UT			LEVEL:	II	DATE:	2024-02-18
INSPECTION EVALUATED BY:	Youhao CO: 320586199812223617			LEVEL:	II	DATE:	2024-02-18

SignatureoftheNDETechnicianisastatementthattheyhavetrainingtotheNDTSpecificationandatteststoinspectionsperformedinaccordancewiththestatedinspection methodology.SupplierNDETechnicianqualificationsandcurrenteyeexaminationsareprovidedandmaintainedbythecompany.

苏州赛普机械有限公司 产品质量检验专用章

Form CNT-005 Rev D

REVIEWED
By Chao Hu at 2:38 pm, Jun 17, 2024

Ultrasonic Inspection Report

Cactus Job#: N/A | Cactus PO#: SUP0025187 | WO #: / | Date: 2024-01-30

Location/Branch: Suzhou Pympe Machinery CO.,Ltd | Material or Type: 4130 75K

Equipment Description: CTS-2020 | Part Number/Rev: 134797/A02 | Material Size/Thickness: Ø282*783

Customer Product Identifier: SUP0025187-2-51~SUP0025187-2-75

Inspection Procedure

NDT Specification: CNT-005,UltrasonicTesting | Revisione: E

Cactus Wellhead NDT Specification contains inspection methodology and acceptance criteria

Mode	☐ Shear ☒ Longitudinal	Scanning Method	☒ Contact ☒ Manual ☐ Immersion ☐ Automatic	Pres. Type	☒ A-Scan ☐ Phasedarray ☐ Digitalthickness	Areas Examined	☒ All accessiblevolume ☐ Pressurecontainingweld(s) ☐ Cladweld(s) ☐ Other(explainbelow)

Surface Condition: After HT ≤250Rms | Material Thickness/Dia: Ø282*783

UT Instrumentmake/model: CTS-2020 | S/N and Cal. Due Date: 526322181062/2024-09-05

Couplant: Machine Oil | Angle: 0°

Cabletype/length: Q9-Q9,Double Q9-C5/1.8m | Transducermanufacturer: Ultrabetter Testing Technology (Suzhou) Co.LTD

Transducersize: Φ20/Φ14/Φ10 | Transducerfrequency: 2.5MHZ/5.0MHZ

Reference dB: 80% | Record Additional dB's For: Curvature: 29dB | Acoustic Differences: 4dB | Scanning: +6dB

(Total): P1-65dB/P2-52dB/P3-62dB/P4-53dB

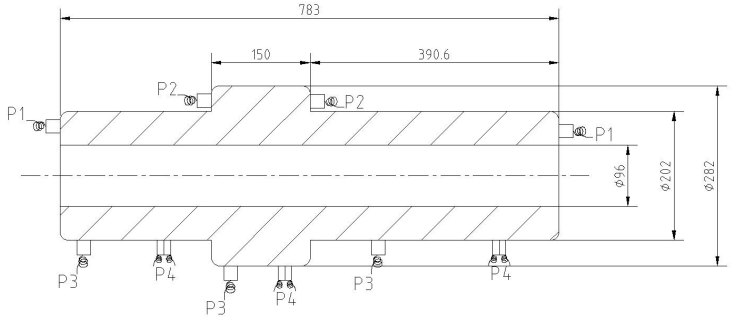
Reference Standards

Block Serial Number	Overall Thickness	Metal Travel Distance	Reflector Size	Reflectororientation
P1:4130-16-0787/4130-16-2362/4130-16-3150	800mm	783mm	1/4"	FBH/straight(2.5Z20N)
P2:4130-8-0099/4130-8-0296/4130-8-0600	152.4mm	150mm	1/8"	FBH/straight(5Z14N)
P3:4130-8-0099/4130-8-0197/4130-8-0394	100mm	93/53mm	1/8"	FBH/straight(5Z14N)
P4:4130-8-0025/4130-8-0059/4130-8-0099	25mm	93/53mm	1/8"	FBH/Double straight(DA5P10FS25)

Inspection Results

Work Order&Serial #	Heat #	Heat Lot #	Serial#(Original)	Accept/Reject Quantities	NCR #(If Applicable)
/	2310345	HY240112-1	SUP0025187-2-51	1 ACCEPT	NA
/	2310345	HY240129-2	SUP0025187-2-52~SUP0025187-2-75	24 ACCEPT	NA

SKETCH AND TECHNIQUE DESCRIPTION (Identify and attach additional drawings/sheets):



Min. Thickness Required:	N/A	Min. Thickness Recorded:	N/A	Accept or Reject:	Accept all		
Inspection Performed By:	Sunpengyang CO:21102429098UT			Level:	II	Date:	2024-01-30
Inspection Evaluated By:	Youhao CO: 320586199812223617			Level:	II	Date:	2024-01-30

SignatureoftheNDETechnicianscertifiestrainingtotheNDTSpecificationandattheteststoinspection methodology.SupplierNDETechnicianqualificationsandcurrenteyeexaminationsareprovidedandmaintainedbythecompany.

REVIEWED
By Chao Hu at 4:26 pm, Jun 14, 2024

REVIEWED
By zhang.xiuwei at 9:18 am, 6/16/24

Magnetic Particle Inspection Report

Cactus

CACTUS JOB #	NA	CACTUS PO #	SUP0025187	SUPPLIER JOB #	\
DATE	2024/04/17	CACTUS LOCATION	CW Suzhou	Mat'l Type or Spec.	4130 125K/CMS-107 H
PART NUMBER/REV:	134797 A02	Product Description	BTM X 30.5 LG X6.125-4 STUB ACME-2G RIGHT HAND BOX TOP & 4-13/16 HBPV THD,SPEC FOR ROTATING CASING STRING,4140 125K,TEMP U,MATL AA,PSL3,PR2		

SURFACE CONDITION ☑ Finish Machined ☐ As forged/as cast ☐ As welded ☐ Prepared for welding/cladding ☐ Plated/coated

AREAS EXAMINED ☐ All accessible wetted/sealing surfaces ☑ All accessible surfaces ☐ Welds ☐ Weld repairs ☐ Weld preps ☐ Other (make comments in remarks column)

INSPECTION PROCEDURE

NDT SPECIFICATION CNT-004 MAGNETIC PARTICLE TESTING **Revision** E

Cactus Wellhead NDT Specification containing inspection methodology and accept/reject criteria.

METHOD	☑ WET FLUORESCENT ☐ DRY COLOR CONTRAST	PARTICLE TYPE & BATCH #	BX-60H 2021-08-03 MAGNAGLO 2023-12-11	BATH STRENGTH	0.3ml/100ml	CURRENT	☑ AC ☐ HWDC ☑ FWDC ☑ FIELD ADEQUACY VERIFIED BY FIELD STRENGTH INDICATOR

TECHNIQUE		Equipment Calibration	ITEM	SERIAL NUMBER	CALIBRATION DUE DATE
☑ CONTINUOUS ☑ MULTIDIRECTIONAL					
☑ CENTRAL CONDUCTOR ☐ HEAD/TAIL SHOT ☑ COIL ☐ CABLE WRAP ☐ LEECHES/CLAMP ☐ YOKE			MAGNETIZING EQUIPMENT	ZYJC20190318	2024-09-04
AMPERAGE/AMP TURNS USED	4220A/2440A/8T		UV-A LIGHT METER	RYWSN010008	2024-06-06
☐ Auto. AC Demag ☑ Manual DC Demag ☐ AC Demag			GAUSS METER	0359	2024-06-04

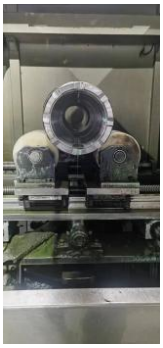
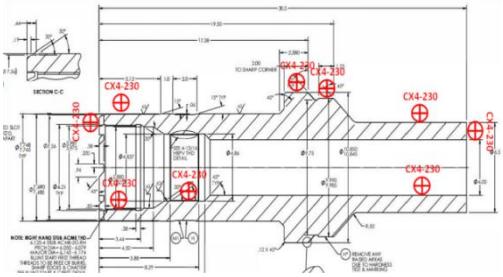
UV-A Light Reading @ 16"	Before use: 2820 µw/cm²	After use (REQUIRED for battery powered light): 2820	µw/cm²
WHITE LIGHT INTENSITY	6 LX	SURFACE TEMPERATURE	☑ VERIFIED SURFACE TEMP IS BELOW 125° F, ACTUAL 71.6° F,
UV-A LAMP USED	☐ MAGNAFLUX EV6000 ☑ UVLED365-SJ4028-6K		

INSPECTION RESULTS

SERIAL NUMBER	HEAT NUMBER	HEAT LOT NUMBER	QTY Inspected	QTY ACCEPT	QTY REJECT	REMARKS
SUP0025187-2-51	2310345	HY240112-1	1	1	\	NA
SUP0025187-2-52~75	2310345	HY240129-2	24	24	\	NA

Comments, location of recordable/rejectable indications, plating/coating type and thickness, etc. (use additional sheet if needed)



INSPECTION PERFORMED BY:	Sunpengyang 21102429098MT 孙鹏扬	LEVEL	MT-2
INSPECTION EVALUATED BY:	XiaoYadong 32002106273MT 肖亚东	LEVEL	MT-3
CUSTOMER REPRESENTATIVE:	NA		

Signature of the NDE Technician verifies training to the NDT Specification and attests to inspections performed in accordance with the stated inspection methodology and results determined to meet the accept/reject criteria of the stated specification. Approved NDE Technician qualifications, certifications, and current eye examinations are maintained.

REVIEWED
By zhang.xiuwei at 7:18 pm, 6/16/24

Magnetic Particle Inspection Report

Cactus

CACTUS JOB # NA **CACTUS PO #** SUP0025187 **SUPPLIER JOB #** \

DATE 2024/04/18 **CACTUS LOCATION** CW Suzhou **MAT'L TYPE OR SPEC.** 4130 125K/CMS-107 H

PART NUMBER/REV: 134797 A02 **PRODUCT DESCRIPTION** BTM X 30.5 LG X6.125-4 STUB ACME-2G RIGHT HAND BOX TOP & 4-13/16 HBPV THD,SPEC FOR ROTATING CASING STRING,4140 125K,TEMP U,MATL AA,PSL3,PR2

SURFACE CONDITION ☑ Finish Machined ☐ As forged/as cast ☐ As welded ☐ Prepared for welding/cladding ☐ Plated/coated

AREAS EXAMINED ☐ All accessible wetted/sealing surfaces ☑ All accessible surfaces ☐ Welds ☐ Weld repairs ☐ Weld preps ☐ Other (make comments in remarks column)

INSPECTION PROCEDURE

NDT SPECIFICATION CNT-004 MAGNETIC PARTICLE TESTING **Revision** E

Cactus Wellhead NDT Specification containing inspection methodology and accept/reject criteria.

METHOD	☑ WET FLUORESCENT ☐ DRY COLOR CONTRAST	PARTICLE TYPE & BATCH #	BX-60H 2021-08-03 MAGNAGLO 2023-12-11	BATH STRENGTH	0.3ml/100ml	CURRENT	☑ AC ☐ HWDC ☑ FWDC ☑ FIELD ADEQUACY VERIFIED BY FIELD STRENGTH INDICATOR

TECHNIQUE		Equipment Calibration	ITEM	SERIAL NUMBER	CALIBRATION DUE DATE
☑ CONTINUOUS ☑ MULTIDIRECTIONAL					
☑ CENTRAL CONDUCTOR ☐ HEAD/TAIL SHOT ☑ COIL ☐ CABLE WRAP ☐ LEECHES/CLAMP ☐ YOKE			MAGNETIZING EQUIPMENT	ZYJC20190318	2024-09-04
AMPERAGE/AMP TURNS USED	4220A/2440A/8T		UV-A LIGHT METER	RYWSN010008	2024-06-06
☐ Auto. AC Demag ☑ Manual DC Demag ☐ AC Demag			GAUSS METER	0359	2024-06-04

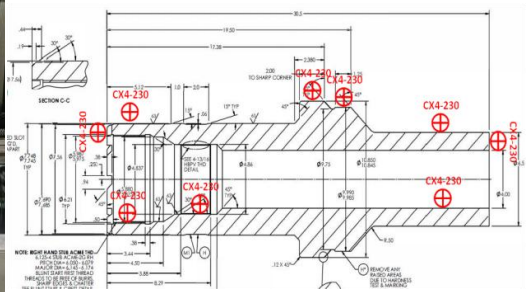
UV-A Light Reading @ 16"	Before use: 2820 µw/cm²	After use (REQUIRED for battery powered light): 2820	µw/cm²
WHITE LIGHT INTENSITY	6 LX	SURFACE TEMPERATURE	☑ VERIFIED SURFACE TEMP IS BELOW 125° F, ACTUAL 75.2 F,
UV-A LAMP USED	☐ MAGNAFLUX EV6000 ☑ UVLED365-SJ4028-6K		

INSPECTION RESULTS

SERIAL NUMBER	HEAT NUMBER	HEAT LOT NUMBER	QTY Inspected	QTY ACCEPT	QTY REJECT	REMARKS
SUP0025187-2-76~100	2310345	HY240129-2	25	25	\	NA

Comments, location of recordable/rejectable indications, plating/coating type and thickness, etc. (use additional sheet if needed)



INSPECTION PERFORMED BY:	Sunpengyang 21102429098MT	LEVEL	MT-2
INSPECTION EVALUATED BY:	XiaoYadong 32002106273MT	LEVEL	MT-3
CUSTOMER REPRESENTATIVE:	NA		

Signature of the NDE Technician verifies training to the NDT Specification and attests to inspections performed in accordance with the stated inspection methodology and results determined to meet the accept/reject criteria of the stated specification. Approved NDE Technician qualifications, certifications, and current eye examinations are maintained.

产品质量证明书
TEST CERTIFICATE

石钢京诚装备技术有限公司

河钢石钢 HBIS SHISTEEL ShiGangJingCheng Equipment Development and Manufacturing

记录编号Record No.：JL/JC-09-223

证书编号 Certificate No.：ZC-23-1523-001

顾客 Customer	苏州派普机械有限公司										合同号 Contract No.		230105XSZC152		炉号 Heat No.		2310345				
技术条件 Specification	QX/SJ.G7090-2019										材质 Steel Grade		AISI4140		数量(件) Pieces		42				
交货状态 Delivery Condition	热轧Hot Rolled					表面质量 Surface quality		合格 OK			规格(mm) size		∮200		重量(t) Weight		66.68				
化学成分 Chemical Composition	C(%)	Si(%)	Mn(%)	P(%)	S(%)	Cr(%)	Cu(%)	Ni(%)	Mo(%)	V(%)	B(%)	H(ppm)	O(ppm)	N(ppm)							
	0.42	0.25	1.00	0.013	0.002	1.00	0.02	0.02	0.220	0.005	0.0005	0.9	9	68							
	-	-	-	-	-	-	-	-	-	-	-	-	-	-							
机械性能 Mechanical Properties	抗拉强度 Tensile Strength Rm(Mpa)	屈服强度 Yield Strength Re(Mpa)	延伸率Elongation A(%)	面缩率Reduction of area Z(%)	冲击功V型(-20℃) Impact AKV（J）	调质硬度QT hardness（HBW）															
	1131	1049	20.0	55	81.8	314															
	1118	1035	18.0	55	85.2	312															
	-	-	-	-	82.4	-															
	-	-	-	-	-	-															

一般疏松 Ordinary Porosity	中心疏松 Center Porosity	一般点状偏析 General segregation	边缘点状偏析 Edge Spot Segregation	非金属夹杂物A细（级） AT (Grade)	非金属夹杂物A粗（级） AH (Grade)	非金属夹杂物B细（级） BT (Grade)	非金属夹杂物B粗（级） BH (Grade)	非金属夹杂物C细（级） CT (Grade)	非金属夹杂物C粗（级） CH (Grade)	非金属夹杂物D细（级） DT (Grade)	非金属夹杂物D粗（级） DH (Grade)	晶粒度 Grain size (Grade)
1.0	1.0	0	0	0.5	0.5	0.5	0	0	0	1.0	0.5	7.0
1.0	1.0	0	0	0.5	0.5	0.5	0	0	0	1.0	0.5	-
-	-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-	-	-

备注 Remar：压缩比Compression ratio：9.00 冶炼工艺方法:BOF+LF+VD+CCM

判定员 Inspector：刘哲　　质量工程师 Engineer：肖鹏

签发日期 Date of issue：2023-08-28

石钢京诚装备技术有限公司 质量证明专用章

地址：辽宁沿海产业基地管委会新联大街东1号

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